

Seeing Only the Winners

Selective Visibility, Congested Routes and Disappointment in Entry Markets

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Abstract

Graduates learn how hard it is to enter a career partly by watching the outcomes of others, and online those outcomes are selectively visible: offers are announced, rejections are not. I study the consequences in a simple model of an entry market with a scarce prestige route and an abundant core route. Applicants who take the visible share of successes at face value overestimate their chances of entering the prestige route. In the unique steady state, more selective visibility raises applications to the prestige route, lowers its acceptance rate, leaves more core vacancies unfilled, and widens the gap between how easy the route looks and how hard it is. The model separates two explanations that look alike in application data. A rise in the status of the prestige route also congests it, but applicants with accurate beliefs anticipate their rejections, so aggregate disappointment, the difference between the offers applicants expect and those they receive, arises only from distorted beliefs, and status magnifies it. The welfare cost of distorted beliefs is bounded above by this disappointment. Informing some graduates of true acceptance rates raises the success rate of all applicants, so accurate information about entry routes is a public good. More generally, when agents learn about opportunities from selectively observed outcomes, the information environment can itself create congestion. I propose a randomised information experiment with final-year students, with a saturation design to measure spillovers, to test the mechanism.

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I Introduction

Competition for graduate jobs in the United Kingdom is at a record high. Members of the Institute of Student Employers received an average of 38 applications per graduate vacancy in 2002/3, 86 in 2022/3 and 140 in each of the two most recent years, the highest since the survey began in 1991, while their graduate hiring fell by 8 per cent in 2024/25 (Institute of Student Employers, 2025). Three explanations dominate discussion of the entry-level market. Firms may be hiring fewer young workers in occupations exposed to artificial intelligence (Brynjolfsson et al., 2026). Language models may have made written applications cheap to produce and therefore uninformative, so that employers struggle to identify able candidates (Galdin and Silbert, 2025). And demand for graduates may simply have weakened with the economic cycle. Each probably matters. This paper studies a further mechanism, located not in how many entry-level jobs exist or how employers screen for them, but in where graduates apply.

Competition is not spread evenly. In 2024 employers in digital and IT received 205 applications per graduate vacancy and those in financial and professional services 188, against 74 in the charity and public sector (Institute of Student Employers, 2024). Two stories about the same market circulate at once: employers in some sectors struggle to fill entry-level roles with good candidates, while good candidates experience the market as closed. A market can feel closed if most applicants queue for a few routes, even when other routes have vacancies.

The mechanism I study concerns how graduates learn which routes are worth queuing for. Offline, a student hears about the outcomes of her close friends, rejections included. Online, she sees the outcomes of hundreds of acquaintances, but only those they choose to announce. An offer from a strategy consultancy or an investment bank is announced; the rejections that preceded it, and the rejections of everyone else who applied, are not. A student who forms her view of her chances from the outcomes she sees will overestimate them. Social media has shifted the sample from which students learn towards the second kind of contact, and its users tend to conclude that others are doing better than they are (Chou and Edge, 2012). Partner markets show the same pattern, and a viral 2024 song in which the singer wants “a man in finance” compressed both markets into a single aspiration.

Standard models of directed search predict that applicants spread themselves across jobs according to the trade-off between pay and the probability of being hired, which they are assumed to know (Moen, 1997). If applicants know the queues, a rise in the attractiveness of a route lengthens its queue only until the lower odds offset the higher payoff. That assumption is the one I relax. A large body of evidence shows that job seekers’ beliefs about their own prospects are biased and respond too little to the truth (Spinnewijn, 2015; Mueller et al., 2021), and that people who learn from others’ outcomes neglect how those outcomes were selected (Enke, 2020; Barron et al., 2024). In entry markets, the selection is built into what is visible.

I model this in the simplest setting I could find in which the mechanism and its consequences can be stated and proved. A cohort of graduates chooses between a prestige route with a fixed number of places and a core route that absorbs everyone who chooses it. Graduates differ only in the value of their core option. Before applying, each graduate sees the outcomes of the previous cohort’s applicants to the prestige route, but successful applicants post their outcome more often than rejected ones. A graduate who treats the share of successes among posts as the success rate commits selection neglect in the sense of Jehiel (2018). In the steady state, beliefs and behaviour are consistent with what each cohort sees.

The model yields five results. First, more selective visibility raises applications to the prestige route and lowers its acceptance rate (Proposition 2); because applicants who are rejected from the prestige route lose

the core route for that cycle, the same change leaves more core vacancies unfilled. Second, it widens the gap between the perceived and the true acceptance rate, so the route looks easier precisely as it becomes harder (Proposition 3). Third, it separates belief distortion from status. A rise in the status of the prestige route also congests it and raises rejections, but with accurate beliefs rejections are anticipated. Aggregate disappointment, the number of offers applicants expect minus the number they receive, is zero whatever the route's status when beliefs are accurate, and positive and increasing in status when they are not (Proposition 4). Status alone does not produce disappointment, but it magnifies the disappointment that distorted beliefs produce. Fourth, the welfare cost of distorted beliefs, over and above the congestion that occurs even with accurate beliefs, is bounded above by aggregate disappointment, and equals half of it when core values are uniform (Proposition 5). Fifth, informing some graduates of the true acceptance rate lowers congestion and raises the success rate of every applicant, informed or not, so accurate information about entry routes is a public good (Proposition 6).

The mechanism is an instance of a more general idea in information economics. When agents learn about opportunities from selectively observed outcomes, the information environment can itself create congestion: survivorship in what is observed becomes crowding in what is pursued. The crowding is endogenous, because the more agents pursue an opportunity, the lower its true success rate, and what the next cohort observes reflects that decline only in part. The same logic applies wherever entry is rationed and success is more visible than failure, as in start-ups, creative careers, academic job markets and partner markets. In the language of information design (Bergemann and Morris, 2019), the visibility of failure is a property of the information structure through which applicants learn. It is therefore something that platforms shape and that policy can change.

The contribution is deliberately narrow. The model takes congestion, social learning and biased beliefs, each well studied on its own, and shows how they combine in entry markets to produce congestion, an expectation gap and a welfare loss that can be ranked against each other. Its sharpest prediction, that belief distortion and status are observationally similar in application data but differ in disappointment, suggests a direct test. Section 5 proposes a randomised experiment with final-year students that elicits beliefs about acceptance rates and corrects them, with a contrast that isolates selection neglect and a saturation design that measures spillovers.

1.1 Related literature

The paper relates first to work on biased beliefs in job search. Spinnewijn (2015) shows that unemployed workers overestimate how quickly they will find work, and Mueller et al. (2021) find that job seekers' beliefs are systematically biased and respond too little to their actual prospects, so that optimistic job seekers with poor prospects select into long-term unemployment. These papers concern beliefs about the rate of finding any job. I study beliefs about the chance of entering a particular route, and derive the bias from what applicants observe rather than assuming it. On the policy side, Belot et al. (2019) show in a field experiment that low-cost advice suggesting alternative occupations broadens job search and raises interviews, especially for job seekers who search narrowly. In the model, such advice works by correcting beliefs about the routes applicants already favour and about the ones they overlook.

Second, the mechanism is a form of selection neglect. Denrell (2003) argues that learning from surviving organisations undersamples failure and produces myths about what succeeds. Jehiel (2018) formalises a related logic in an equilibrium model in which investors who learn from the returns of implemented projects become overoptimistic and overinvest, and Barron et al. (2024) confirm the prediction experimentally. Enke

(2020) shows experimentally that people neglect information they do not see, even when they know it exists. I adapt the analogy-based approach of Jehiel (2018) to a market in which the selection lies in what is posted rather than in what is implemented, and in which a fixed number of places makes the error costly to others. The resulting over-entry recalls the excess entry that Camerer and Lovallo (1999) attribute to overconfidence, and the competition for a few positions resembles the winner-take-all markets of Frank and Cook (1995), where over-entry arises even without biased beliefs.

Third, the paper relates to directed search. In Moen (1997), applicants who know the trade-off between wages and queue length internalise congestion, and Banfi and Villena-Roldán (2019) find that job ads offering higher wages attract more applicants, as directed search predicts. Under random rationing, which I assume, applicants who know their odds still ignore that they reduce the odds of others, so the model has a congestion externality even with accurate beliefs. Distorted beliefs add to it. Oyer (2008) shows that stock market conditions at graduation shape entry into investment banking, an example of how aggregate conditions determine which routes graduates attempt.

Fourth, the paper connects to work on social comparison and reference-dependent preferences. Social media worsens the mental health of young users (Braghieri et al., 2022; Allcott et al., 2020), with unfavourable social comparison among the proposed channels, and learning about others' pay lowers the satisfaction of those who earn less (Card et al., 2012; Perez-Truglia, 2020). Genicot and Ray (2017) model aspirations formed by observing others and show that aspirations set too high relative to achievable outcomes can frustrate rather than motivate. I do not model utility from comparison. The disappointment I measure is the gap between the offers applicants expect and those they receive, the input to a reference-dependent loss in the sense of Kőszegi and Rabin (2006). Giuliano and Spilimbergo (2025) review evidence that shocks experienced in young adulthood have strong and lasting effects on preferences and beliefs, which is one reason why disappointment at the point of entry into the labour market may matter beyond the cycle in which it occurs.

Finally, the empirical design builds on information experiments about the returns to education and the choice of major (Jensen, 2010; Wiswall and Zafar, 2015), and on two-stage designs that randomise the saturation of an intervention to measure displacement and spillovers (Crépon et al., 2013; Baird et al., 2018).

The remainder of the paper is organised as follows. Section 2 sets out the mechanism and the model. Section 3 presents the results. Section 4 discusses the application to partner markets. Section 5 sets out testable predictions and an empirical design. Section 6 discusses limitations and Section 7 concludes. Proofs not given in the text are in Appendix A.

2 Model

2.1 The mechanism

The argument has four steps.

- (i) *Selective visibility.* Graduates learn about entry routes partly from the outcomes of earlier applicants, and online, successes are posted more often than failures.
- (ii) *Selection neglect.* Graduates who take the visible share of successes at face value overestimate their chances of entering routes whose successes are most visible.
- (iii) *Congestion.* Because the number of places on those routes is fixed, more applicants means a lower acceptance rate, and graduates who would have done better on another route are rejected instead.
- (iv) *Disappointment.* Applicants receive fewer offers than they expected. The gap between expected and

realised outcomes is the input to negative assessments of the market.

The status of a route is a separate force. It raises the value of success and therefore the number of applicants, but it does not by itself create a gap between expected and realised outcomes. The model is built so that the two forces can be distinguished.

2.2 Environment

A cohort of graduates of unit mass chooses between two routes. The *prestige route* P has a mass $k \in (0, 1)$ of places and gives value $V > 0$ to a graduate who obtains one. V includes both pay and the status of the route. The *core route* C absorbs every graduate who chooses it, which isolates congestion on the prestige route; Section 6 discusses congested core routes. Graduate i obtains value c_i on the core route, where c_i is drawn independently from a distribution F . Each graduate applies to exactly one route in a cycle. This is the simplest way to give the prestige route an opportunity cost: applications to it take time and effort, and a graduate who pursues it and fails ends the cycle worse off than one who took the core route. A graduate rejected from the prestige route obtains zero, normalising the value of a lost cycle; the remark at the end of Section 3 shows that the cost need not be total.

If a mass n of graduates apply to the prestige route, places are rationed at random, so each applicant is hired with probability

$$q(n) = \min\{1, k/n\}. \quad (1)$$

Random rationing is a reduced form of screening with noisy signals among applicants who are similar on what employers observe. It excludes sorting by ability, which I discuss in Section 6.

A graduate who believes she will be hired with probability $\hat{q}$ applies to the prestige route if and only if $\hat{q}V \geq c_i$. Since F is continuous, ties have probability zero.

Assumption 1 (Primitives). F is continuous and strictly increasing on its support $[0, \bar{c}]$, and $k < F(V) < 1$.

The condition $k < F(V)$ says that the prestige route would be oversubscribed if success were certain. The condition $F(V) < 1$ says that some graduates prefer the core route even then. For closed forms I use a uniform benchmark in which $F(c) = c$ on $[0, 1]$ and $V \in (k, 1)$.

The core route has a mass $m \geq 1 - k$ of vacancies. Vacancies are filled by graduates who choose the core route; graduates rejected from the prestige route cannot fill them in the same cycle. The number of unfilled core vacancies is therefore $U = m - (1 - n)$.

2.3 Beliefs under selective visibility

Graduates form beliefs before applying by observing the outcomes of the previous cohort's applicants to the prestige route. Each successful applicant posts her outcome with probability $\sigma_S > 0$, and each rejected applicant posts hers with probability $\sigma_F = \lambda \sigma_S$, where $\lambda \in (0, 1]$ is the *relative visibility of failure*. With $\lambda = 1$, failures are as visible as successes.

Social media lowers λ mainly by changing whose outcomes a graduate observes and remembers, rather than by making people more boastful. People present themselves positively offline as well as online (Masciantonio et al., 2025), and in private conversation they often hide their successes from peers who are struggling (Prinsloo et al., 2026). Offline, however, a graduate learns about the outcomes of close ties, whose rejections she hears about. Online, she observes many weak ties (Granovetter, 1973), and learns their outcomes only when they choose to announce them, which they do for offers far more than for rejections. As weak ties

make up more of her sample, the relative visibility of failure falls. Consistent with this, [Chou and Edge \(2012\)](#) find that Facebook users with more friends they do not know personally agree more strongly that others have better lives.

A graduate observes many posts and estimates the success rate as the share of successes among them. She neglects that failures are under-posted. This is the analogy-based reasoning of [Jehiel \(2018\)](#): she uses the empirical frequency of the outcomes she observes without correcting for how they were selected.

Lemma 1 (Selective visibility). *If the previous cohort's acceptance rate is q , a graduate who observes a large number of posts estimates the success rate as*

$$b(q; \lambda) = \frac{q}{q + \lambda(1 - q)}, \quad \text{equivalently} \quad \frac{b}{1 - b} = \frac{1}{\lambda} \cdot \frac{q}{1 - q}. \quad (2)$$

For $q \in (0, 1)$, b is strictly increasing in q and strictly decreasing in λ , and $b(q; \lambda) > q$ if and only if $\lambda < 1$.

Proof. Successes are posted at rate $q\sigma_S$ and failures at rate $(1 - q)\lambda\sigma_S$, so by the law of large numbers the share of successes among posts converges to $b(q; \lambda)$. The odds form follows directly. The monotonicity properties follow from the odds form, since the odds are strictly increasing in q and strictly decreasing in λ , and $b > q$ if and only if $1/\lambda > 1$. ■

Selective visibility therefore inflates the perceived odds of success by the factor $1/\lambda$. A graduate who corrected for selection would recover q exactly.

Remark. The belief $\hat{q}$ is a graduate's belief about her own chances, which can differ from her belief about the average applicant. Suppose she scales the visible odds by a factor $\theta > 0$, so that her belief has odds $(\theta/\lambda) q/(1 - q)$. Overconfidence about one's own prospects ([Spinnewijn, 2015](#)) corresponds to $\theta > 1$ and compounds selective visibility. The evidence that social media leaves users feeling worse off than others ([Chou and Edge, 2012](#); [Braghieri et al., 2022](#)) cuts both ways. It is consistent with an inflated view of how others fare, which is what $\lambda < 1$ captures, but it could also lower a graduate's view of her own standing, $\theta < 1$. All results below hold with λ replaced by λ/θ whenever $\lambda/\theta < 1$, that is, whenever the inflation of others' success outweighs any deflation of one's own standing. Social comparison can also raise V : if entering the prestige route is what one's visible peers appear to achieve, failing to enter it carries a positional cost.

2.4 Equilibrium

I study a steady state in which each cohort forms beliefs from the posts of the previous one and the acceptance rate is constant across cohorts.

Definition 1. A *steady state* is a mass of applicants n^* and an acceptance rate q^* such that $q^* = q(n^*)$, every graduate holds the belief $\hat{q}^* = b(q^*; \lambda)$, and $n^* = F(V\hat{q}^*)$.

The benchmark with accurate beliefs is $\lambda = 1$, in which $\hat{q}^* = q^*$. I denote it by n^R and q^R .

Proposition 1 (Existence and uniqueness). *Under Assumption 1, there is a unique steady state, and $k < n^* < F(V)$. In the uniform benchmark,*

$$n^* = \frac{-(1 - \lambda)k + \sqrt{(1 - \lambda)^2 k^2 + 4\lambda V k}}{2\lambda} \quad (\lambda < 1), \quad n^R = \sqrt{V k}. \quad (3)$$

The prestige route is always oversubscribed, so the acceptance rate $q^* = k/n^*$ lies strictly between zero and one. As $\lambda \rightarrow 0$, perceived success approaches certainty and $n^* \rightarrow F(V)$.

Remark. The steady state is the limit of the learning process. In the uniform benchmark, suppose that each cohort applies according to beliefs formed from the previous cohort's posts, so that $n_{t+1} = Vb(q(n_t); \lambda)$. For any $\lambda < 1$ and any initial $n_0 > 0$, n_t converges to n^* , with damped oscillations: a heavily oversubscribed cohort posts few successes, which discourages the next cohort, and so on (Lemma 2 in the appendix). With $\lambda = 1$ and backward-looking beliefs the process cycles, as in a cobweb model. The benchmark with accurate beliefs should therefore be read as one in which graduates know the current acceptance rate.

3 Results

3.1 Congestion

Proposition 2 (Congestion). *The mass of applicants n^* is strictly decreasing in λ and strictly increasing in V . The acceptance rate q^* is strictly increasing in λ and strictly decreasing in V .*

More selective visibility therefore draws more graduates to the prestige route and lowers its acceptance rate. The effect is self-limiting but does not undo itself: the lower acceptance rate is visible in the next cohort's feed and dampens the distortion, but the fixed point still lies above the benchmark. A rise in status has the same qualitative effect on applications, which is why application data alone cannot separate the two forces.

The congestion has a counterpart elsewhere in the market. Two quantities that are often cited as evidence of opposite problems move together.

Corollary 1 (Two stories at once). *The mass of graduates rejected from the prestige route, $R = n^* - k$, and the mass of unfilled core vacancies, $U = m - (1 - n^*)$, are both strictly decreasing in λ .*

As visibility becomes more selective, more graduates fail to obtain the job they aimed for and more employers on the core route fail to fill their vacancies. Neither requires any change in the number of jobs.

3.2 The perception gap

Proposition 3 (Perception gap). *In the steady state, $\hat{q}^* \geq q^*$, with equality if and only if $\lambda = 1$. The perceived acceptance rate $\hat{q}^*$ is strictly decreasing in λ , so the gap $\hat{q}^* - q^*$ is strictly decreasing in λ .*

Proof. The first claim follows from Lemma 1 since $q^* \in (0, 1)$. Because $n^* = F(V\hat{q}^*)$ and F is strictly increasing on its support, $\hat{q}^* = F^{-1}(n^*)/V$, which is strictly increasing in n^* and hence, by Proposition 2, strictly decreasing in λ . Since q^* is strictly increasing in λ , the gap is strictly decreasing in λ . ■

Equilibrium feedback partly offsets the distortion, since a lower true acceptance rate is what the next cohort sees. Proposition 3 shows that the offset is never complete. As failure becomes less visible, the route looks easier from the outside while becoming harder to enter. Figure 1(a) illustrates with $k = 0.1$ and $V = 0.6$ in the uniform benchmark. With accurate beliefs, 24.5 per cent of the cohort applies and 41 per cent of applicants succeed. If failures are posted at a quarter of the rate of successes ($\lambda = 0.25$), 36 per cent apply, 28 per cent succeed, and applicants believe their chance is 60 per cent.

3.3 Status and beliefs

Let

$$D \equiv n^* (\hat{q}^* - q^*) = n^* \hat{q}^* - k \quad (4)$$

denote *aggregate disappointment*. Applicants as a group expect $n^* \hat{q}^*$ offers and receive k , so D is the number of expected offers that do not materialise. Measured in units of value, VD is the aggregate shortfall of realised against anticipated payoffs among applicants. I do not model how disappointment shapes assessments of the market. I treat it as the input that any such mapping would use: under reference-dependent preferences it is the loss relative to expectations (Kőszegi and Rabin, 2006), and for a graduate who learns about the market from her own experience it is the surprise that moves her beliefs. Social comparison may also raise the weight attached to it, since a rejected graduate who believes her peers are faring better experiences the shortfall as falling behind and not only as a missed offer.

Proposition 4 (Status versus beliefs).

- (i) The mass of rejected applicants $R = n^* - k$ is strictly increasing in V and strictly decreasing in λ .
- (ii) If $\lambda = 1$, then $D = 0$ for every V .
- (iii) If $\lambda < 1$, then $D > 0$, D is strictly increasing in V , and D is strictly decreasing in λ .

Proposition 4 separates the two forces. A rise in the status of the prestige route congests it and raises the number of rejected graduates, but if graduates know their odds, their rejections are anticipated: each applicant took a gamble she understood. Disappointment in the aggregate requires distorted beliefs. Status nevertheless matters for disappointment, because it magnifies the effect of distorted beliefs: the more attractive the route, the more graduates whose beliefs are inflated decide to apply. In the uniform benchmark, the two forces combine in a single expression,

$$D = \frac{1-\lambda}{\lambda} k (1 - \hat{q}^*), \quad (5)$$

which is zero when $\lambda = 1$ and, since $\hat{q}^*$ falls as V rises, increasing in V when $\lambda < 1$. Figure 1(b) illustrates.

This is the sense in which a rise in status can still produce disappointment: a status shift on its own does not, but a status shift in a market where failure is invisible does. It also suggests the discriminating test. Status and selective visibility both predict more applications to prestige routes, but only selective visibility predicts that applicants' beliefs about their odds exceed the true odds.

3.4 Welfare

Welfare is the total realised value of the cohort, evaluated at true outcomes. Graduates who obtain a place receive V , those who choose the core route receive their core value, and those who are rejected receive zero. Since applicants are those with core values below $F^{-1}(n)$, welfare when a mass $n \geq k$ applies is

$$W(n) = kV + \int_{F^{-1}(n)}^{\bar{c}} c dF(c), \quad (6)$$

which is strictly decreasing in n . A planner who could choose who applies would send exactly the k graduates with the lowest core values, so $W(k)$ is the first best. Welfare losses can be decomposed as

$$W(k) - W(n^*) = \underbrace{[W(k) - W(n^R)]}_{\text{congestion externality}} + \underbrace{[W(n^R) - W(n^*)]}_{\text{belief distortion } L}. \quad (7)$$

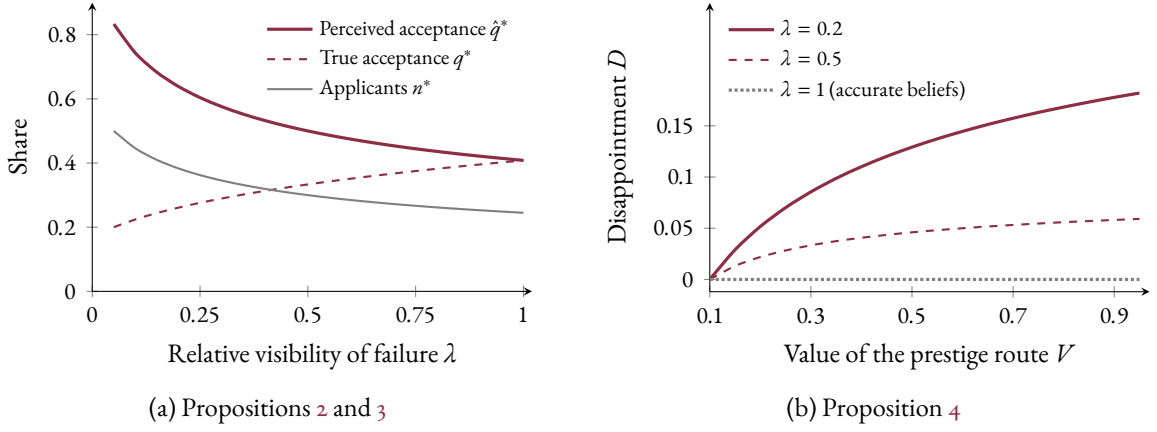


Figure 1: Selective visibility, congestion and disappointment in the uniform benchmark with $k = 0.1$. Panel (a): as failure becomes less visible, more graduates apply to the prestige route, its acceptance rate falls and its perceived acceptance rate rises ($V = 0.6$). Panel (b): with accurate beliefs, a rise in the value of the prestige route produces no aggregate disappointment; with distorted beliefs, it produces more.

The first term is positive even with accurate beliefs: under random rationing, each applicant who knows her odds ignores that she lowers the odds of others, as in models of rent-seeking contests and winner-take-all markets (Frank and Cook, 1995). The second term is the cost of selective visibility.

Proposition 5 (Welfare).

- (i) Steady-state welfare $W(n^*)$ is strictly increasing in λ .
- (ii) The welfare cost of distorted beliefs satisfies

$$L = VD - \int_{Vq^R}^{V\hat{q}^*} F(c) dc \leq VD, \quad (8)$$

with strict inequality when $\lambda < 1$.

- (iii) In the uniform benchmark, $L = VD/2$.

The welfare cost of distorted beliefs is therefore bounded above by aggregate disappointment measured in units of value. The bound is intuitive. Disappointment counts the full shortfall between anticipated and realised payoffs, whereas the welfare cost counts only the core value that the marginal applicants gave up; applicants whose core value was low lost little by trying. The bound makes disappointment a conservative measure of what distorted beliefs cost, at least in terms of realised payoffs. If disappointment itself reduces wellbeing, as under reference-dependent preferences, the cost is larger.

3.5 Information provision

Suppose a share $\pi \in [0, 1]$ of graduates observe the true acceptance rate, for instance because published applications-per-place statistics reach them, while the rest form beliefs from posts. The steady state then solves

$$n = \pi F(Vq(n)) + (1 - \pi) F(Vh(q(n); \lambda)). \quad (9)$$

Proposition 6 (Information as a public good). Let $\lambda < 1$. For each $\pi \in [0, 1]$ there is a unique steady state. The mass of applicants $n^*(\pi)$ is strictly decreasing in π and the acceptance rate $q^*(\pi)$ is strictly increasing in π . In particular, informing some graduates raises the success probability of every applicant, including those

who remain uninformed. Holding the acceptance rate fixed, an informed graduate applies with strictly lower probability than an uninformed one.

Correcting one graduate's beliefs benefits the other applicants, through shorter queues, and the graduates on the core route who would otherwise have been drawn away from it. A graduate who acquires information captures only part of its value, so information about entry routes is under-provided. This is consistent with Belot et al. (2019), where cheap occupational advice raised interviews, and with the observation that core vacancies go unfilled when graduates are not told how many there are. In the uniform benchmark, numerical computation over a wide grid of parameters shows that welfare also rises and aggregate disappointment falls as π increases; I have not proved these properties in general.

Remark. None of the results depends on the normalisation that rejected applicants obtain zero. Suppose that a rejected applicant obtains δc_i with $\delta \in [0, 1)$, because she can enter the core route late or in a worse match. Graduates then apply if and only if $c_i \leq \tau(\hat{q}) \equiv \hat{q}V/[1 - \delta(1 - \hat{q})]$. The qualitative statements in Propositions 1 to 4 and Proposition 6 use only the fact that the application threshold is continuous and strictly increasing in $\hat{q}$, so they continue to hold; the welfare expressions change. The same is true if graduates are risk averse.

4 Partner markets

The model can be read as a stylised description of partner markets on dating platforms. The prestige route is pursuit of a partner in the top tier of desirability, k is the capacity of that tier to respond, and the core route is pursuit of partners whose desirability is closer to one's own. Visible outcomes are again selected. Couples are visible on social media, while unanswered messages and declined matches are not.

There is evidence consistent with each step. Using messaging data from a large dating site, Bruch and Newman (2018) find that men and women both contact partners who are on average about 25 per cent more desirable than themselves, and that the probability of a reply falls steeply as the gap in desirability widens. Hitsch et al. (2010) document strong and largely shared preferences over partners' attributes on an online dating site, which concentrate demand on the same people. The model predicts that greater visibility of successful matches with highly desirable partners raises the volume of messages to the top tier, lowers reply rates and widens the gap between expected and realised replies, with the same distinction between status and belief distortion as in Proposition 4.

The two markets are also linked. Part of the value V of a prestige career is its value in the partner market, which is what the "man in finance" slogan expresses. If the visibility of successful couples raises the status of prestige careers, a distortion in the partner market raises V in the labour market, and by Proposition 4 it magnifies labour market disappointment where beliefs are already distorted.

The analogy has limits. Partner markets are two-sided: the top tier chooses whom to answer, so places are not rationed at random, and the capacity to respond is not fixed. The model treats the top tier as passive. A two-sided version would let the top tier respond to the volume of messages, for instance by raising its standards, which would lower acceptance rates further.

5 Empirical strategy

5.1 Predictions

The ideal experiment would randomly remove a group of students from social media at the age when they form views about which careers to pursue, and compare their applications, beliefs and later assessments of the labour market with those of students who remained. Deactivation experiments of this kind exist for other outcomes (Allcott et al., 2020), but one long and targeted enough for career choice is impractical. The model instead suggests tests that work through beliefs, which can be measured and randomly corrected.

Hypothesis 1 (Belief gap). Students' perceived acceptance rates for prestige routes exceed the true rates, and the gap is larger for students whose information about outcomes comes more from weak ties online.

Hypothesis 2 (Response to information). Correcting beliefs about acceptance rates reduces applications to prestige routes and raises applications to core routes, by more for students and routes with larger initial gaps. Students whose beliefs are already accurate do not change their applications, however highly they value the route.

Hypothesis 3 (Disappointment). Correcting beliefs reduces the shortfall of offers relative to expectations and makes assessments of the graduate labour market less negative, even if it does not increase the number of offers received.

Hypothesis 4 (Spillovers). Raising the share of informed students in a pool of applicants raises the acceptance rate faced by uninformed students in that pool.

Hypothesis 5 (Visibility across routes). Across routes, the growth in applications per vacancy is larger where successful outcomes are more visible online, conditional on changes that affect all routes, such as tools that make applications cheaper to produce.

Hypotheses 1 and 3 follow from Propositions 3 and 4, Hypotheses 2 and 4 from Proposition 6 and Lemma 1, and Hypothesis 5 from Proposition 2 applied to routes that differ in λ . The second part of Hypothesis 2 is the individual-level counterpart of Proposition 4: a student who applies because she values the route highly, knowing her odds, should not respond to information about those odds. Hypothesis 3 is stated without conditioning on outcomes, since offers are themselves affected by treatment.

5.2 A randomised information experiment

Sample and timing. The experiment would recruit final-year undergraduates who intend to apply for graduate roles, through careers services at around ten UK universities, in September and October, before the main application deadlines. Follow-up surveys take place at the end of the recruitment season and about a year after graduation. The design should be pre-registered, with primary outcomes, hypotheses and specifications fixed in advance, and approved by an ethics committee. Every treatment provides only accurate information, and there is no deception.

Baseline. For a fixed list of routes r , the baseline survey elicits each student's perceived probability of an offer if she applied, $\hat{q}_{ir0}$, and her belief about the acceptance rate of the average applicant. Beliefs about the average applicant can be checked against published statistics and are incentivised with a binarised scoring rule (Hossain and Okui, 2013). Beliefs about her own chances cannot be checked at that point and are not

incentivised. The list contains prestige routes, such as strategy consulting and investment banking, and core routes with published statistics, such as the Civil Service Fast Stream, NHS graduate schemes and large-employer schemes in engineering and logistics. The survey also measures how attractive the student finds each route if entry were guaranteed, V_{ir} , which separates status from perceived odds. It also measures where her information about career outcomes comes from: how many people she knows to have received an offer or a rejection for each route, and how many of them she knows personally. Hypothesis 1 is tested on these baseline data before any treatment.

Treatments. Students are randomly assigned with equal probability to four arms, stratified by university, subject group, gender and tercile of baseline belief gap.

Arm	Content
C : Placebo	Application timelines for the same routes, matched in length and format.
T_1 : Base rates	Applications per place and acceptance rates for each route, from published sources.
T_2 : Selection	The same numbers framed as a correction for selection: for each offer announcement a student is likely to see for a route, roughly how many applicants were rejected.
T_3 : Alternatives	Vacancies and acceptance rates for core routes only, in the spirit of Belot et al. (2019).

Each comparison answers a separate question. T_1 against C estimates the effect of accurate information about acceptance rates. T_2 against T_1 holds the numbers fixed and varies only whether the information is framed as a correction for unseen failures, which tests selection neglect directly (Enke, 2020). T_3 against C estimates the effect of information about the core route alone, separating information about where to apply from information that corrects beliefs about the prestige route.

Outcomes. Primary outcomes are pre-specified. They are (i) the posterior belief gap $\hat{q}_{ir1} - q_r$, elicited at the end of the baseline session; (ii) applications to each route, recorded in the follow-up survey and, for students who consent, verified against careers-service records; (iii) the shortfall of offers relative to expectations, $D_i = \sum_r A_{ir} \hat{q}_{ir1} - O_i$, where A_{ir} indicates an application and O_i is the number of offers; and (iv) an index of assessments of the graduate labour market. Secondary outcomes include offers, employment and job satisfaction a year after graduation, measures of self-esteem and morale, and an optional module on the partner market.

Estimation. Intention-to-treat effects are estimated by

$$Y_i = \alpha + \sum_{j=1}^3 \beta_j T_{ij} + \gamma Y_{i0} + X_i' \delta + \eta_{s(i)} + \varepsilon_i, \quad (10)$$

where Y_{i0} is the baseline value of the outcome where one exists, X_i are pre-specified covariates and $\eta_{s(i)}$ are strata fixed effects. The sharpest test of the mechanism uses variation within students across routes. In the stacked student-by-route data,

$$A_{ir} = \mu_i + \nu_r + T_i \times (\kappa G_{ir0} + \kappa_V G_{ir0} V_{ir} + \rho V_{ir}) + Z_{ir}' \zeta + u_{ir}, \quad (11)$$

where T_i indicates T_1 or T_2 , $G_{ir0} = \hat{q}_{ir0} - q_r$ is the baseline belief gap, μ_i and ν_r are student and route fixed effects, and Z_{ir} contains G_{ir0} , V_{ir} and their product. The main effect of treatment is absorbed by μ_i . Since a strong candidate may rightly believe her chances exceed the average, the same specification is estimated with the gap in beliefs about the average applicant, which is free of this concern. In the model, correcting a belief by $\Delta\hat{q}$ shifts the application threshold by $V\Delta\hat{q}$, so Hypothesis 2 predicts $\kappa < 0$ and $\kappa_V < 0$: treatment reduces applications most for the routes on which a student was most optimistic, and by more the more she values them. It also predicts $\rho = 0$: for a route on which the student's beliefs were accurate, treatment has no effect however highly she values it. Using $T_i \times G_{ir0}$ as an instrument for the posterior belief $\hat{q}_{ir1}$ in an equation for A_{ir} gives the response of applications to perceived odds, which corresponds to the slope of the application threshold in the model. The exclusion restriction requires that treatment affects applications only through beliefs about acceptance rates; re-eliciting V_{ir} after treatment checks that information did not also change perceived status.

Threats to validity. Experimenter demand is limited by a placebo of equal length, by measuring behavioural outcomes weeks after treatment in a separate survey, and by bounding demand effects (de Quidt et al., 2018). Attrition from follow-up is reduced with incentives, tested for balance by arm, and addressed with bounds (Lee, 2009). Primary outcomes are grouped into indices with false discovery rate adjustment (Anderson, 2008). Self-reported applications are verified where records allow. Students talk to each other, so individual randomisation within a university biases estimates of direct effects towards zero; the saturation design below turns this problem into an object of study.

Power. With 3,000 students and 750 per arm, the minimum detectable effect for a pairwise comparison is about 0.15 standard deviations at 80 per cent power and a 5 per cent two-sided test, before covariate adjustment. Controlling for baseline beliefs, which are likely to be strongly correlated with posterior beliefs, lowers this further, and the within-student design in (II) uses variation across routes.

5.3 Extension: a saturation design for spillovers

Proposition 6 predicts that information has effects beyond those who receive it: by shortening queues, it raises the acceptance rate faced by uninformed applicants. Testing this requires varying the share of informed students in a pool of applicants, which a two-stage randomised design does (Crépon et al., 2013; Baird et al., 2018).

Design. Clusters are subject cohorts within universities. In the first stage, each cluster is assigned a saturation $s \in \{0, 0.25, 0.75\}$, with clusters at $s = 0$ serving as pure controls. In the second stage, students in each cluster are assigned to an information arm with probability s and to the placebo otherwise. The design identifies three effects: the direct effect of information at saturation s , comparing treated and untreated students in the same cluster; the spillover on untreated students, comparing them with students in pure control clusters; and the total effect of treating a share s of a cluster. Inference is at the cluster level, with randomisation inference as a check, and power depends mainly on the number of clusters.

Two channels. Spillovers on the untreated can arise in two ways. Information can diffuse, so that untreated students' beliefs move towards the truth. Or queues can shorten, so that untreated students face higher acceptance rates even if their beliefs do not change. Measuring beliefs and outcomes separately for untreated

students distinguishes them: a spillover on success rates without a spillover on beliefs is the signature of the congestion channel in Proposition 6.

Where congestion can be detected. National graduate schemes draw applicants from many universities, so even a fully treated cluster is a small share of their pool and the congestion channel will be undetectable there. It can be detected only where the pool of applicants is confined to the cluster, for instance placement-year schemes allocated within a university, departmental or university-specific internship schemes, or employer events with a fixed number of places per campus. The spillover analysis should be pre-registered for these opportunities, while diffusion of beliefs can be measured for all routes.

5.4 An observational test

Hypothesis 5 can be examined with existing data. Employer surveys report applications per vacancy by sector over time (Institute of Student Employers, 2025), and the visibility of successful outcomes by sector can be proxied by the frequency of posts announcing offers on professional and social platforms. A difference-in-differences comparison of sectors with high and low visibility, around the growth of short-form video platforms, would distinguish the mechanism from explanations that operate across all routes. Signal collapse (Galdin and Silbert, 2025) and cheaper applications raise applications everywhere and need not create a gap between perceived and true acceptance rates. Selective visibility predicts that both the growth in applications and the belief gap are concentrated where success is most visible.

6 Discussion

The model has several limitations, each of which suggests an extension.

First, rationing is random, so the model has no role for ability. If employers screen on ability, the graduates who are rejected are disproportionately those with weaker applications, and a rejected graduate faces a further inference problem: whether rejection reflects her ability or the length of the queue. Selective visibility sharpens this problem. By Bayes' rule, the higher the acceptance rate a graduate believed she faced, the more her rejection appears to reveal about her own ability, so the same rejection damages her self-assessment more when failure is invisible. A graduate who cares about her self-image may instead attribute the rejection to the market (Kőszegi, 2006). This is the natural route to modelling effects on self-esteem, which the present model does not capture.

Second, each graduate makes a single application, and the core route takes everyone. In practice, graduates apply to a portfolio of reaches and safeties (Chade and Smith, 2006), and tools that make applications cheaper raise the number they make, which may congest core routes too: in 2025, retail, consumer goods and tourism employers received the most applications per graduate hire (Institute of Student Employers, 2025). With multiple applications, distorted beliefs would tilt portfolios towards prestige routes. If graduates also believe that core routes, whose outcomes almost no one posts, are near-certain safeties, they will make too few safety applications and some will end the cycle with nothing. A rejection from a route thought to be safe is the largest surprise, so disappointment per rejection would be greatest on the least prestigious routes, and by the argument above such rejections would also do most damage to self-assessment. The interaction with signal collapse (Galdin and Silbert, 2025) also becomes relevant: the longer the queue for a route, the harder the screening problem for its employers.

Third, the number of places is fixed and employers do not respond. Employers on oversubscribed routes might raise their standards or expand intake, and employers on the core route might raise pay to attract

graduates. Endogenous wages would bring the model closer to directed search, in which posted terms partly offset congestion.

Fourth, the model is a steady state in which each cohort learns afresh from the previous one and rejected graduates leave the market. In practice, rejected graduates apply again, so repeat applicants accumulate and add to congestion. They also learn from their own rejections, but if they update slowly (Mueller et al., 2021), they keep aiming high, which lengthens spells without work and delays their entry into the core route. A dynamic version would therefore speak to graduate unemployment and to the efficiency of matching between employers and graduates, and since graduates rarely post what they learn, their correction would still not reach the next cohort. A related extension would let some graduates correct for selection, as in Jehiel (2018), who shows that rational investors can amplify the overoptimism of naive ones. Here, sophisticated graduates behave like the informed graduates of Proposition 6.

Fifth, the model takes the route choice as given at graduation. If entering a prestige route requires earlier investments, such as the choice of degree, beliefs formed from selective visibility would shape those investments too, and the distortion would propagate upstream, which may help explain why degree choices concentrate in particular subjects.

Finally, I do not model how disappointment becomes an assessment of the market, or to whom rejected graduates attribute their failure. Proposition 4 identifies disappointment as the quantity that distinguishes belief distortion from status, and Giuliano and Spilimbergo (2025) suggest why experiences at labour market entry may shape beliefs for a long time. How disappointment is attributed, to oneself, to employers, to the economy or to technology, is a separate question that the present model leaves open.

7 Conclusion

Online, entry into careers and relationships is visible mainly through those who succeed. This paper has shown, in a simple model, what follows when applicants learn from such selected outcomes. Applications concentrate on routes whose successes are most visible, those routes become harder to enter while appearing easier, other routes are left with unfilled vacancies, and applicants as a group receive fewer offers than they expected. A rise in the status of prestige routes congests them too, but produces disappointment only in combination with distorted beliefs. The welfare cost of distorted beliefs is bounded by that disappointment, and accurate information about entry routes, by shortening queues for everyone, is a public good.

The mechanism is one example of a more general point: when people learn about opportunities from selectively observed outcomes, the information environment itself can create congestion. It is one of several mechanisms that may explain why many graduates experience the entry-level labour market as closed. Its distinctive prediction is a gap between perceived and true acceptance rates that is largest where success is most visible, and that prediction can be tested directly by measuring and randomly correcting students' beliefs.

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A Proofs

Throughout, write $b(n) \equiv b(q(n); \lambda)$ when λ is fixed, and recall from Assumption 1 that $F(V) < 1$ implies $V < \bar{c}$, so that $Vb \in (0, \bar{c})$ whenever $b \in (0, 1]$ and F is strictly increasing at Vb .

Proof of Proposition 1. For $n \leq k$, $q(n) = 1$ and $b(1; \lambda) = 1$, so the right-hand side of the steady-state condition equals $F(V) > k \geq n$, and there is no steady state with $n \leq k$. On $[k, 1]$, define

$$\varphi(n) \equiv n - F(Vb(k/n; \lambda)).$$

φ is continuous. Since k/n is strictly decreasing in n , b is strictly increasing in q (Lemma 1) and F is non-decreasing, $F(Vb(k/n; \lambda))$ is non-increasing in n , so φ is strictly increasing. Moreover, $\varphi(k) = k - F(V) < 0$ and $\varphi(1) = 1 - F(Vb) \geq 1 - F(V) > 0$. Hence φ has a unique root $n^* \in (k, 1)$, and $n^* = F(V\hat{q}^*) < F(V)$ because $\hat{q}^* = b(k/n^*; \lambda) < 1$ for $n^* > k$ and F is strictly increasing on $[0, \bar{c}]$.

In the uniform benchmark, the steady-state condition is $n = Vb(k/n; \lambda)$. Substituting $q = k/n$ into (2) gives $b = k/[k + \lambda(n - k)]$, so the condition becomes $\lambda n^2 + (1 - \lambda)kn - Vk = 0$. For $\lambda \in (0, 1)$, the product of the roots is $-Vk/\lambda < 0$, so exactly one root is positive, and it is the expression in (3). For $\lambda = 1$, $n^2 = Vk$. ■

Proof of Proposition 2. Write $\varphi(n; \lambda, V)$ for the function in the proof of Proposition 1. For $n \in (k, 1)$, $b(k/n; \lambda)$ is strictly decreasing in λ (Lemma 1), and F is strictly increasing at Vb , so φ is strictly increasing in λ . Let $\lambda' > \lambda$. Then $\varphi(n^*(\lambda); \lambda', V) > \varphi(n^*(\lambda); \lambda, V) = 0$. Since $\varphi(\cdot; \lambda', V)$ is strictly increasing in n and has its unique root at $n^*(\lambda')$, it follows that $n^*(\lambda') < n^*(\lambda)$. The same argument applies to V , since φ is strictly decreasing in V for $n \in (k, 1)$. The statements about $q^* = k/n^*$ follow immediately. ■

Proof of Corollary 1. Both R and U are strictly increasing functions of n^* , which is strictly decreasing in λ by Proposition 2. ■

Proof of Proposition 4. (i) $R = n^* - k$, and the claim follows from Proposition 2. (ii) If $\lambda = 1$, then $\hat{q}^* = b(q^*; 1) = q^*$, so $D = 0$. (iii) If $\lambda < 1$, $\hat{q}^* > q^*$ by Proposition 3, so $D > 0$. For the dependence on V , fix $\lambda < 1$ and write $D = \psi(n^*) - k$, where

$$\psi(n) \equiv n b(k/n; \lambda) = \frac{nk}{k + \lambda(n - k)}, \quad \psi'(n) = \frac{(1 - \lambda) k^2}{(k + \lambda(n - k))^2} > 0.$$

Since λ is fixed, D depends on V only through n^* , which is strictly increasing in V by Proposition 2. Hence D is strictly increasing in V . For the dependence on λ , write $D = n^* \hat{q}^* - k$. Both n^* and $\hat{q}^*$ are positive and strictly decreasing in λ (Propositions 2 and 3), so their product is strictly decreasing in λ .

For (5), the uniform steady-state condition $\lambda n^2 + (1 - \lambda)kn - Vk = 0$ gives $n^2 - Vk = (1 - \lambda)k(V - n)/\lambda$. Since $\hat{q}^* = n^*/V$ in the uniform benchmark, $D = n^* \hat{q}^* - k = (n^{*2} - Vk)/V = (1 - \lambda)k(1 - \hat{q}^*)/\lambda$. ■

Proof of Proposition 5. By (6), $W'(n) = -F^{-1}(n) < 0$ for $n \in (k, 1)$, so W is strictly decreasing in n , and (i) follows from Proposition 2.

For (ii), let $c^R \equiv F^{-1}(n^R) = Vq^R$ and $c^* \equiv F^{-1}(n^*) = V\hat{q}^*$ denote the thresholds below which graduates apply, with $c^R \leq c^*$ by Proposition 2. By (6) and integration by parts,

$$L = W(n^R) - W(n^*) = \int_{c^R}^{c^*} c dF(c) = c^* F(c^*) - c^R F(c^R) - \int_{c^R}^{c^*} F(c) dc.$$

Now $c^* F(c^*) = V\hat{q}^* n^*$ and $c^R F(c^R) = Vq^R n^R = Vk$. Hence $c^* F(c^*) - c^R F(c^R) = V(n^* \hat{q}^* - k) = VD$, which gives (8). When $\lambda < 1$, $c^* > c^R > 0$ and $F > 0$ on (c^R, c^*) , so the integral is strictly positive.

For (iii), with $F(c) = c$ on $[0, 1]$, $\int_{c^R}^{c^*} F(c) dc = \int_{c^R}^{c^*} c dF(c) = L$, so $L = VD - L$ and $L = VD/2$. ■

Proof of Proposition 6. For $n \leq k$, the right-hand side of (9) equals $F(V) > k \geq n$, so any steady state has $n > k$. On $[k, 1]$ define

$$\chi(n; \pi) \equiv n - \pi F(Vk/n) - (1 - \pi)F(Vb(k/n; \lambda)).$$

As in the proof of Proposition 1, χ is continuous and strictly increasing in n , with $\chi(k; \pi) = k - F(V) < 0$ and $\chi(1; \pi) \geq 1 - F(V) > 0$, so there is a unique root $n^*(\pi) \in (k, 1)$. Moreover,

$$\frac{\chi}{\pi} = F(Vb(k/n; \lambda)) - F(Vk/n) > 0 \quad \text{for } n \in (k, 1),$$

since $b(q; \lambda) > q$ for $\lambda < 1$ and $q \in (0, 1)$ (Lemma 1) and F is strictly increasing on $[0, \bar{c}]$. By the argument in the proof of Proposition 2, $n^*(\pi)$ is strictly decreasing in π , and so $q^*(\pi) = k/n^*(\pi)$ is strictly increasing. Every applicant is hired with probability $q^*(\pi)$ whether informed or not. Finally, at any acceptance rate $q \in (0, 1)$, an informed graduate applies with probability $F(Vq)$ and an uninformed one with probability $F(Vb(q; \lambda)) > F(Vq)$. ■

Lemma 2 (Stability). *In the uniform benchmark, let $n_{t+1} = f(n_t) \equiv Vb(q(n_t); \lambda)$ for $t \geq 0$, with $n_0 > 0$. If $\lambda < 1$, then $n_t \rightarrow n^*$. If $\lambda = 1$, then $f(f(n)) = n$ for all $n \in [k, V]$, so the process cycles unless it starts at n^R .*

Proof. For $n \leq k$, $f(n) = V$. For $n \geq k$, substituting $q = k/n$ into (2) gives

$$f(n) = \frac{Vk}{\lambda n + (1 - \lambda)k},$$

which is strictly decreasing. Let $I \equiv [f(V), V]$. Since $V > k$, $f(V) = Vk/[\lambda V + (1 - \lambda)k] \geq k$, since $(1 - \lambda)(V - k) \geq 0$; so $I \subset [k, V]$. Since f is decreasing on $[k, \infty)$ and $f(k) = V$, f maps $[k, V]$ into I . Hence $n_1 \leq V$; and $n_2 \in I$, because either $n_1 \leq k$, in which case $n_2 = V$, or $n_1 \in (k, V]$, in which case $n_2 \in I$. So $n_t \in I$ for all $t \geq 2$.

On I , f is the Möbius transformation associated with the matrix

$$M = \begin{pmatrix} 0 & Vk \\ \lambda & (1 - \lambda)k \end{pmatrix},$$

whose fixed points correspond to the eigenvectors of M . The eigenvalues μ_1, μ_2 satisfy $\mu_1\mu_2 = -\lambda Vk < 0$ and $\mu_1 + \mu_2 = (1 - \lambda)k$. If $\lambda < 1$, then $\mu_1 + \mu_2 > 0$, so $\mu_1 \neq -\mu_2$ and $\mu_1^2 \neq \mu_2^2$. Then M^2 has distinct eigenvalues and the same eigenvectors as M , so $g \equiv f \circ f$ has the same fixed points as f : n^* and a negative number. On I , therefore, g is continuous, strictly increasing, maps I into itself and has the unique fixed point n^* . A continuous increasing self-map of a compact interval with a unique fixed point generates monotone sequences that converge to it, so both (n_{2t}) and (n_{2t+1}) converge to n^* , and so does (n_t) .

If $\lambda = 1$, then $\mu_1 + \mu_2 = 0$, so $M^2 = \mu_1^2 I$ is a multiple of the identity and g is the identity on $[k, V]$; directly, $f(n) = Vk/n$ and $f(f(n)) = n$. ■